Supplementary Material

The transition to a sustainable economy is not sustainable: an ecological macroeconomic analysis of climate policy, the circular economy, and global inequality

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Appendix 1: MVP Model Equations

The model consists of difference equations describing the dynamics of the economic variables over a finite time horizon, both at the industry level and at the aggregate level. Each period corresponds to one year. Variables are expressed at time t, unless a "-1" subscript denotes a lagged value. The superscript z denotes a region ($Z_1 = \text{EU}$, $Z_2 = \text{RoW}$); f denotes the foreign region. Operators $\odot$ and $\oslash$ denote element-wise (Hadamard) multiplication and division; standard matrix multiplication is denoted by $\cdot$.

Block 1. Households

Real household consumption:

$$c^z = \alpha_1^z \frac{YD_w^z}{p_A^z} + \alpha_2^z \frac{YD_c^z}{p_A^z} + \alpha_3^z \frac{V_{-1}^z}{p_{A,-1}^z} \tag{1.1}$$

where p_A^z is the consumer price index and $\alpha_1^z > \alpha_2^z > \alpha_3^z$ are the propensities to consume out of disposable labour income (YD_w^z), disposable capital income (YD_c^z), and net wealth (V^z), respectively.

Disposable income:

$$YD^z = WB^z + DIV^z + FB^{z+} + r^z_{m,-1} M^z_{h,-1} + r^z_{b,-1} B^z_{s,z,-1} + xr^f_{-1} r^f_{b,-1} B^f_{s,z,-1} + \Delta xr^f B^f_{s,z,-1} + E^{f+}_{s,z,-1} - r^z_{h,-1} L^z_{h,-1} - T^z \tag{1.2}$$

Net wealth accumulation: $V^z = V^z_{-1} + YD^z - p^z_A c^z$ (1.3).

Block 2. Production Firms (Current Account)

The final demand faced by production firms is made up of household consumption, corporate investment in fixed capital, government spending, and public investment. Considering $K = 54$ industries at the global level, the demand for final goods and services across both regions stacks into a $2K$-dimensional vector:

$$\mathbf{d} = \sum_{z \in \{Z_1, Z_2\}} \left(\beta^z c^z + \sigma^z g^z + \iota^z i^z_d + \iota^z_g \, i^z_{d,g} \right) \tag{2.1}$$

where i^z_d is real corporate demand for investment, g^z is real government consumption, β^z is the vector of household consumption shares ($\sum_{s=1}^{2K} \beta^z_s = 1$), ι^z is the vector of investment shares ($\sum_{s=1}^{2K} \iota^z_s = 1$), σ^z is the vector of government spending shares ($\sum_{s=1}^{2K} \sigma^z_s = 1$), and ι^z_g is the vector of public investment shares.

Equivalently, in compact matrix–vector form:

$$\mathbf{d} = \Phi \, \mathbf{f} \tag{2.1$'$}$$

where $\Phi = \left[\beta^{Z_1}, \beta^{Z_2}, \sigma^{Z_1}, \sigma^{Z_2}, \iota^{Z_1}, \iota^{Z_2}, \iota^{Z_1}_g, \iota^{Z_2}_g\right] \in \mathbb{R}^{2K \times 8}$ is the demand-share matrix and $\mathbf{f} = \left(c^{Z_1}, c^{Z_2}, g^{Z_1}, g^{Z_2}, i^{Z_1}_d, i^{Z_2}_d, i^{Z_1}_{d,g}, i^{Z_2}_{d,g}\right)^\top \in \mathbb{R}^8$ collects all regional expenditure aggregates.

Each share vector $\beta^z, \sigma^z, \iota^z \in \mathbb{R}^{2K}$ spans all $2K$ sectors across both regions, so that cross-border imports are embedded within the allocation shares rather than separated into explicit export/import terms. Off-diagonal column sums of Φ give bilateral import flows directly; no separate net-export terms are required.

Gross output via the Leontief inverse:

$$\mathbf{x} = (I - A)^{-1} \, \mathbf{d} = L \, \mathbf{d} \tag{2.2}$$

Corporate profit (residual): $\Pi^z = Y^z - WB^z - r^z_{l,-1}L^z_{f,-1} - \delta K^z_{-1}$ (2.5).

Block 3. Production Firms (Capital Account)

Target capital stock: $K^{z*} = \kappa^z \mathbf{p}^{z\top}\mathbf{x}^z$ (3.1).

Real investment adjustment: $i^z_d = i^z_{d,-1} + \gamma^z(K^{z*} - K^z_{-1})/p^z_{K,-1}$ (3.2).

Capital stock evolution: $K^z = K^z_{-1}(1-\delta^z) + p^z_K i^z_d$ (3.4).

Block 4. Commercial Banks

Loans supplied on demand: $L^z_s = L^z_d$ (4.1). Deposits supplied on demand: $M^z_s = M^z_d$ (4.2).

Block 5. Government

Government expenditure is exogenous. Deficit: $GDEF^z = G^z + r^z_{b,-1}B^z_{s,-1} - T^z - F^z_{cb}$ (5.1). Bills outstanding: $B^z_s = B^z_{s,-1} + GDEF^z$ (5.2).

Block 6. Ecological Extensions

Total CO_2 emissions: $\varepsilon^z = \varepsilon^{z\top}\mathbf{x}^z$ (6.1). Total material extraction: $M^z = \mathbf{m}^{z\top}\mathbf{x}^z$ (6.2). Land use: $\Lambda^z = \lambda^{z\top}\mathbf{x}^z$ (6.3). Water use: $\Omega^z = \omega^{z\top}\mathbf{x}^z$ (6.4).

CE Scenario Design

Final-demand shocks shift fraction ρ of primary-sector demand toward the secondary sector. First-order change in primary material output for material r:

$$\Delta M^r_1 = \rho \cdot fd_{r,1} \cdot (l_{r1,r2} - l_{r1,r1}) \tag{SD.1}$$

Intermediate-production shocks adjust the A-matrix: $\Delta a_{r1,sj} = -\rho\, a_{r1,sj}$, $\Delta a_{r2,sj} = +\rho\, a_{r1,sj}$. First-order material impact:

$$\Delta M^r_1 = \rho \cdot x_{r,1} \cdot (l_{r1,r2} - l_{r1,r1}) \tag{SD.2}$$

Since $(l_{r1,r2} - l_{r1,r1}) \approx -1$ across all material pairs, the ranking of CE interventions by primary material savings reduces to a ranking of the scale factor ($fd_{r,1}$ or $x_{r,1}$).

Appendix 2: Sector List

The model comprises 54 sectors per region (108 total across EU and RoW). Each region shares an identical sector structure. Secondary/reprocessing sectors involved in circular economy substitutions are marked with $\star$.

Appendix 3: Model Calibration

The model is calibrated to 2018 EXIOBASE 3 data, aggregated from 44 countries and 5 rest-of-world blocks to two regions (EU and RoW) and from 163 sectors to 54 using concordance tables. Target calibration values are shown in Table A2.

Table 2: Target calibration values (€10 billion). Source: EXIOBASE 3.

Variable	EU (Z_1)	RoW (Z_2)
Consumption	626.32	2370.71
Investment	245.77	1135.84
Government spending	239.46	675.33
Final exports	102.78	66.36
Final imports	66.36	102.78
GDP	1130.67	4162.75
Gross output	2153.60	8424.52

The remaining parameters (propensities to consume, portfolio coefficients, mark-up rates, speed-of-adjustment parameters, ecological intensity vectors) are calibrated via an evolutionary random-search algorithm. Starting from initial conditions, the algorithm searches over instrument variables to minimize the goodness-of-fit criterion:

$$\mathcal{G} = \sum_{v} \left(\frac{\hat{y}_v - y_v^*}{y_v^*} \right)^2$$

where $\hat{y}_v$ is the model-simulated value and y_v^* is the EXIOBASE target for variable v. At each iteration i, instrument values are perturbed from a Gaussian distribution with mean θ_{i-1} and standard deviation σ (a hyperparameter); the new set is accepted if $\mathcal{G}_i < \mathcal{G}_{i-1}$. The algorithm

Table 1: Sector List. CE substitution pairs by transmission regime. F = Final Demand, I = Intermediate Demand. Primary = source sector; Secondary = reprocessing sector.

#	Sector	Role	Scenario	Channel	Demand	Transmission
1	Agriculture					
2	Animal Farming					
3	Forestry, Logging					
4	Fishing					
5	Fossil-Fuel Extraction					
6	Mining					
7	Meat	Primary	Sc1	Cons.	F	Prod. Leakage
8	Other Food	Secondary	Sc1	Cons.	F	Prod. Leakage
9	Tobacco					
10	Textiles					
11	Wood	Primary	Sc3, Sc7	Cons., Prod.	F, I	Sym. Contraction
12	Wood Re-processing	Secondary	Sc3, Sc7	Cons., Prod.	F, I	Sym. Contraction
13	Pulp	Primary	Sc8	Prod.	I	Sym. Contraction
14	Pulp Re-processing	Secondary	Sc8	Prod.	I	Sym. Contraction
15	Paper					
16	Fossil-Fuel Processing					
17	Plastics	Primary	Sc4, Sc9	Cons., Prod.	F, I	Prod. Leakage
18	Plastics Re-processing	Secondary	Sc4, Sc9	Cons., Prod.	F, I	Prod. Leakage
19	Chemicals					
20	Rubber, Plastic Products					
21	Glass	Primary	Sc11	Prod.	I	Sym. Contraction
22	Glass Re-processing	Secondary	Sc11	Prod.	I	Sym. Contraction
23	Other Non-metallic Mineral Products					
24	Cement	Primary	Sc12	Prod.	I	Sym. Contraction
25	Clinker Re-processing	Secondary	Sc12	Prod.	I	Sym. Contraction
26	Metals	Primary	Sc6, Sc10	Fix. Inv., Prod.	F, I	Comp. Displacement
27	Metals Re-processing	Secondary	Sc6, Sc10	Fix. Inv., Prod.	F, I	Comp. Displacement
28	Durable Goods					
29	Furniture					
30	Recycling					
31	Fossil-Fuel Electricity	Primary	Sc2, Sc13	Cons., Prod.	F, I	Fossil Collapse
32	Renewable Electricity	Secondary	Sc2, Sc13	Cons., Prod.	F, I	Fossil Collapse
33	Energy Transmission, Distribution					
34	Gas					
35	Water					
36	Construction	Primary	Sc5, Sc14	Fix. Inv., Prod.	F, I	Const. Rebound
37	Construction Re-processing	Secondary	Sc5, Sc14	Fix. Inv., Prod.	F, I	Const. Rebound
38	Vehicle Sale, Maintenance, Repair					
39	Vehicle Fuel Sale					
40	Retail, Wholesale Trade					
41	Leisure					
42	Transport					
43	Professional Services					
44	Real Estate					
45	Machine, Equipment Renting					
46	R&D					
47	Public Administration, Defence					
48	Education					
49	Health, Social Work					
50	Waste Incineration					
51	Biogas, Composting					
52	Waste Water					
53	Waste Landfill					
54	Other Services					

terminates when $\mathcal{G}$ reaches a satisfactory level, typically below 0.01 for all 14 target variables simultaneously. Initial balance-sheet configurations (asset and liability stocks) are included among the instrument variables to ensure stock-flow consistency from period one.

Appendix 4: List of Tables and Figures

Tables

Figures